

1. How fast would you have to move a **0.50 m** wire perpendicular to Earth's magnetic field, which is **$5.0 \times 10^{-5} \text{ T}$** , to induce an EMF of **1.5 V**?
2. An airplane with a wingspan of **12 m** travels at right angles through a magnetic field of strength **$5.0 \times 10^{-3} \text{ T}$** . An EMF of **1.5 V** is induced between the ends of the wingtips. How fast was the airplane moving?
3. The armature of a **60 Hz** AC generator rotates in a magnetic field of strength **0.48 T**. If the area of the coil is **$2.4 \times 10^{-2} \text{ m}^2$** and it contains **120 loops**, what will the peak EMF be?

